

EDISON'S BRIGHT IDEA

A year after Swan demonstrated his bulb, the American inventor Thomas Edison (see pp.186–187) came up with his own version, using a filament made of carbonized (charred) bamboo.

Edison ensured its popularity by also devising the electricity-generating power plants and distribution systems that powered the lights, bringing electricity into millions of people's homes for the first time.



MODERN BULBS

Today, many incandescent light bulbs (which create light by heating a filament—a wire—in the bulb) are being replaced in homes with more energy-efficient options. These include compact fluorescent bulbs (see p.183) and light-emitting diodes (LEDs), such as the ones shown above. Some LED lights can be controlled with a smart device, such as a phone, linked to a central control hub.

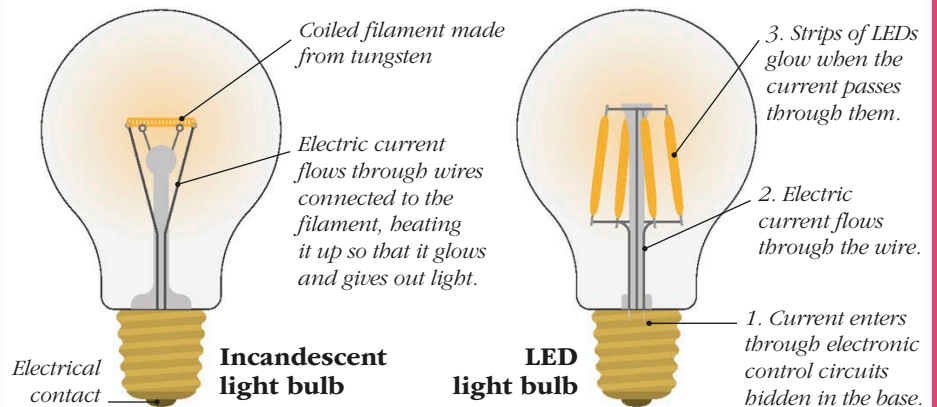
AT HOME

Almost total vacuum inside bulb

Loop-shaped filament

Connecting wires deliver electric current.

COMPARING BULBS



When you flick the switch, an electric current passes through the bulb's filament, making it glow. Because the bulb contains nonreactive gases instead of oxygen, the filament doesn't burn out quickly.

An LED is made of a material called a semiconductor. When you flick the switch, charged particles called electrons move through this material, creating energy in the form of light.

◀ JOINING FORCES

This early Edison light bulb dates from 1879. Despite initial disagreements over patents for their invention, Swan and Edison joined forces in 1883 to form the Edison and Swan United Electric Company, later called Ediswan, in Britain.